RBE 3002 Lab 4 Report

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Introduction:

In lab four, we combine what we learned throughout the term in order to enable our robot to navigate and explore an unknown maze. We were able to implement a SLAM routine capable of identifying and traversing a map using A* path planning and PI controlled locomotion while continually avoiding obstacles. Additionally, we were tasked with solving the kidnapping problem, in which our robot is placed in a random spot on the map and must identify its current position using AMCL and navigate to a designated area on the map.

Methodology:

Our lab was split into three main phases: exploration, localization, and navigation. Like we accomplished in the last lab, we needed to calculate the configuration space of the map. By going through every grid and appending it to a list of occupied cells or of unknown cells. From there, we can identify the frontier based on its neighbors. We calculated the average of the frontier in order to find the centroids, which we used to navigate the robot to explore the field. We then used our A* algorithm from the last lab to find the most optimized path—where the goal is the centroids—and we navigate through the map until there is nothing left to explore. Once that occurs, the robot homes to its starting position, finishing up both phase 1 and 2.

After generating the map, the robot will then be placed in a random location on the map. From there, it must localize and determine where it is as well as navigate to a specific destination. The starting position and the destination are both decided by course staff. The final phase is done 3 times, and the best time from the starting cell to the goal location is used.

Results:

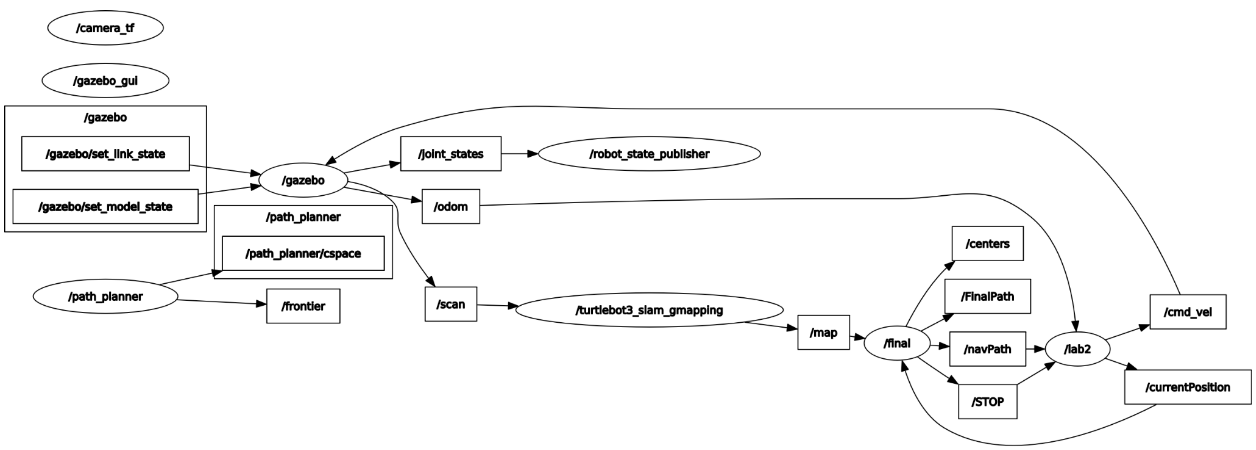

Figure 1 - System architecture for phase 1 & 2

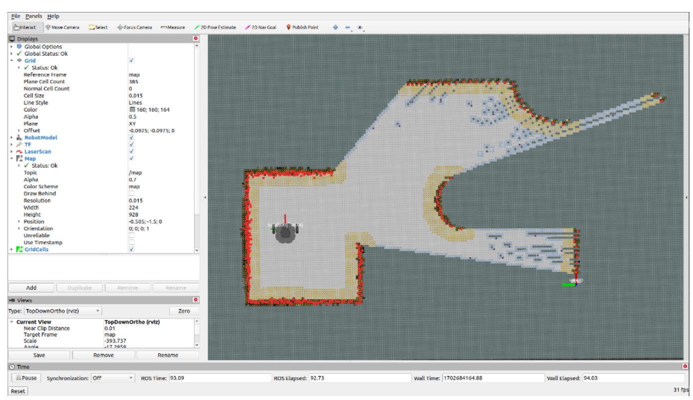

Figure 2 - Frontier generation and configuration-space calculation with red representing the LIDAR, orange representing the c-space, black representing the occupied cells, and blue representing frontier cells

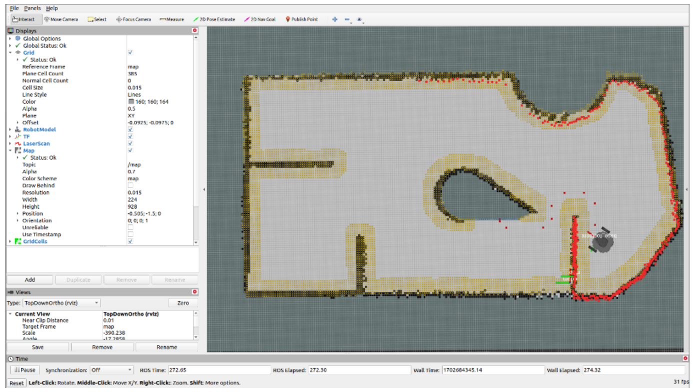

Figure 3 - Frontier exploration using A* path planning with the same configuration as the previous figure. Red grid cells in the free space represent waypoints for path traversal

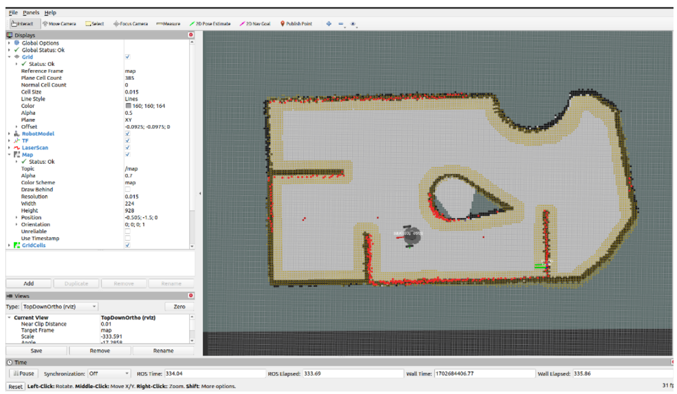

Figure 4 - Home Navigation after successful mapping

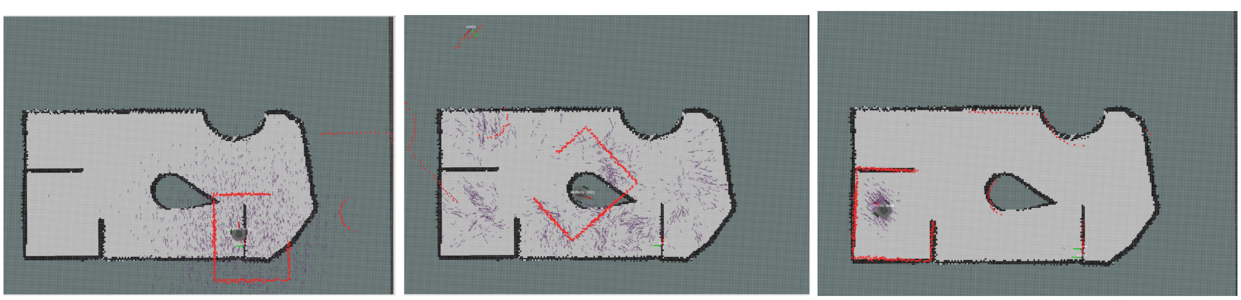

Figure 5-7 - Simulated AMCL before (left), during (middle), and after (right) localization with the purple arrows representing the particle cloud of AMCL

```xml
<!-- AMCL -->
<node pkg="amcl" type="amcl" name="amcl">
  <param name="min_particles"             value="500"/>
  <param name="max_particles"             value="3000"/>
  <param name="kld_err"                   value="0.02"/>
  <param name="update_min_d"              value="0.10"/> <!--from 0.2-->
  <param name="update_min_a"              value="0.05"/> <!--from 0.2-->
  <param name="resample_interval"         value="1"/>
  <param name="transform_tolerance"       value="0.5"/>
  <param name="recovery_alpha_slow"       value="0.00"/>
  <param name="recovery_alpha_fast"       value="0.00"/>
  <param name="initial_pose_x"            value="$(arg initial_pose_x)"/>
  <param name="initial_pose_y"            value="$(arg initial_pose_y)"/>
  <param name="initial_pose_a"            value="$(arg initial_pose_a)"/>
  <param name="gui_publish_rate"          value="50.0"/>

  <remap from="scan"                      to="$(arg scan_topic)"/>
  <param name="laser_max_range"           value="3.5"/>
  <param name="laser_max_beams"           value="180"/>
  <param name="laser_z_hit"               value="0.5"/>
  <param name="laser_z_short"             value="0.05"/>
  <param name="laser_z_max"               value="0.05"/>
  <param name="laser_z_rand"              value="0.5"/>
  <param name="laser_sigma_hit"           value="0.2"/>
  <param name="laser_lambda_short"        value="0.1"/>
  <param name="laser_likelihood_max_dist" value="2.0"/>
  <param name="laser_model_type"          value="likelihood_field"/>

  <param name="odom_model_type"           value="diff-corrected"/>
  <param name="odom_alpha1"               value="0.1"/>
  <param name="odom_alpha2"               value="0.1"/>
  <param name="odom_alpha3"               value="0.1"/>
  <param name="odom_alpha4"               value="0.1"/>
  <param name="odom_frame_id"             value="odom"/>
  <param name="base_frame_id"             value="base_footprint"/>
</node>
```

Figure 8 - The parameters within ROS AMCL.

Discussion:

Throughout the first phase, we needed to consider many things. While we could build off of the previous code in labs 2 and 3, there were still scenarios we needed to consider. Especially when it came to a c-space potentially being an occupied cell, our A* algorithm had its flaws. We needed to ensure that if that were the case, the robot would not accidentally navigate into an obstacle. We accounted for this while calculating our centroids by having the robot go to the free space near the centroid rather than the centroid itself as shown in Figure 2. Additionally, we needed to have a way for the robot to choose the most efficient frontier centroid. We did so through having the heuristic consider both the length of the frontier and the length of the path. The largest frontier would ensure more exploration, and the length of the path would consider the closest one through the wavefront algorithm. It would then add the closest frontier to a priority queue. Additionally, we tweaked our robot's locomotion to have PID control. Once there are no more frontiers, the robot will return back to the starting position, seen in Figure 4, completing phase 2 and ready for the final phase.

Phase 3 needed the implementation of adaptive monte carlo localization. We considered many of the ROS AMCL parameters, and Figure 8 shows our changes. Figures 5, 6, and 7 show the stages of the robot's localization. It is only after the covariance values of the particle clouds reach a certain value that the robot idles until we give it a navigation goal, completing the kidnapped robot problem.

Conclusion:

By the end of the term, we successfully implemented autonomous navigation onto the turtlebot. We are proficient in ROS and python as well as knowledgeable in various search algorithms, configuration space calculations, and localization algorithms.

References:

Petrović, Luka. *Motion planning in high-dimensional spaces*. 2018. University of Zagreb, Laboratory for Autonomous Systems and Mobile Robotics (LAMOR). https://arxiv.org/pdf/1806.07457.pdf

Topiwala, Anirudh; Inani Pranav; Kathpal, Abhishek. *Frontier Based Exploration for Autonomous Robot*. 2018. University of Maryland. https://arxiv.org/pdf/1806.03581.pdf

Quin, Phillip; Alempijevic, Alen; Paul, Gavin; Liu, Dikai. *Expanding Wavefront Frontier Detection: An Approach for Efficiently Detecting Frontier Cells*. 2014. University of Technology Sydney, Australia. https://opus.lib.uts.edu.au/bitstream/10453/30533/1/quinACRA2014.pdf'

Contributions:

The contributions listed represent our split of the work done throughout the final project. All members actively participated and were responsible for key sections of our final project.

Member	Riley Blair	Jai Jariwala	Tracy Yang
Percentage	50%	25%	25%